THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Assertion (A): Ignoring air resistance in a model of a falling ball is acceptable for a stone. Reason (R): For a stone, the effect of air resistance is very small compared with its weight. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Milk shows the Tyndall effect but salt solution does not. Reason (R): The particles of milk are larger than the particles of a true solution. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): The law of constant proportions is true for compounds but not for mixtures. Reason (R): In a compound the elements are chemically combined in a fixed ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Most alpha particles pass undeflected through a thin gold foil. Reason (R): The nucleus occupies a very small fraction of the atom's volume. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): A body can have zero displacement but non-zero distance. Reason (R): Distance is the length of the path and displacement is the net change in position. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Assertion (A): A cricketer moves his hands backward while catching a fast ball. Reason (R): This increases the time of stopping and hence decreases the force on the hands. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Assertion (A): No work is done when a person pushes against a rigid wall. Reason (R): The displacement of the wall is zero. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): Sound cannot travel in a vacuum. Reason (R): Sound waves need a material medium for propagation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): Red blood cells swell and burst in distilled water. Reason (R): Water enters the cells by osmosis because they lack a cell wall. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): Grasses grow again after being cut. Reason (R): Intercalary meristem is present at the base of their leaves. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): Whale is placed in the class Mammalia. Reason (R): It has mammary glands and breathes with lungs. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Plants raised by vegetative propagation are genetically identical to the parent. Reason (R): They are produced by mitosis from a single parent. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): The polar regions are colder than the equator. Reason (R): Sunlight falls at a slant at the poles and is spread over a larger area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. The mass of the Earth is about 6 x 10^24 kg and that of the Sun about 2 x 10^30 kg. The Sun is about ___ times heavier. 1

(a) 3 x 10^4 **(b)** 3 x 10^6 **(c)** 3 x 10^54 **(d)** 3 x 10^5

15. A solution has 30 g of solute in 120 g of solvent. 90 g of the solvent is evaporated. The new percentage by mass of solute is: 1

(a) 25 per cent **(b)** 50 per cent **(c)** 75 per cent **(d)** 20 per cent

16. A compound of A and B contains 75 per cent A by mass. A second compound of the same elements contains 60 per cent A. The ratio of the masses of A that combine with a fixed mass of B in the two compounds is: 1

(a) 4 : 3 **(b)** 3 : 2 **(c)** 2 : 1 **(d)** 3 : 1

17. An ion has 18 electrons and a nucleus with 16 protons. The ion is: 1

(a) $\mathrm{Cl^-}$ **(b)** Ar **(c)** $\mathrm{S^{2+}}$ **(d)** $\mathrm{S^{2-}}$

18. A body is thrown upward with velocity u. The velocity at half the maximum height is: 1

(a) u / 4 **(b)** u / 2 **(c)** u / sqrt(2) **(d)** u

19. A loaded truck of mass 4000 kg and an empty truck of mass 1000 kg are pushed by equal forces. The ratio of their accelerations (loaded : empty) is: 1

(a) 2 : 1 **(b)** 4 : 1 **(c)** 1 : 4 **(d)** 1 : 1

20. A body is thrown upward with KE of 200 J. At half the maximum height the KE is (ignoring air resistance): 1

(a) 50 J **(b)** 200 J **(c)** 100 J **(d)** 0

21. A sound wave travels from air (340 m/s) into water (1500 m/s). A 500 Hz wave has wavelengths in air and water of: 1

(a) 3 m and 0.68 m **(b)** 0.68 m and 0.68 m
(c) 3 m and 3 m **(d)** 0.68 m and 3 m

22. A cell is placed in a solution and its mass increases by 20 per cent. The solution is: 1

(a) hypertonic **(b)** saturated salt
(c) hypotonic to the cell **(d)** isotonic

23. A ring of bark is removed from a branch. After some days, the part above the ring swells due to: 1

(a) accumulation of food **(b)** minerals
(c) gas **(d)** accumulation of water

24. A new organism is found to be multicellular, with cell walls of chitin, non-motile, and with thread-like hyphae. Its kingdom is: 1

(a) Protista **(b)** Plantae **(c)** Animalia **(d)** Fungi

25. A woman's menstrual cycle is 28 days and menstruation begins on 1 March. The likely day of ovulation is: 1

(a) 28 March **(b)** 7 March **(c)** 1 March **(d)** 14 March

26. A surface of albedo 0.5 receives 800 W/m squared. The absorbed radiation per square metre in 10 s is: 1

(a) 8000 J **(b)** 800 J **(c)** 4000 J **(d)** 400 J

27. A scientist says 'The model fits in 99 of 100 trials, so the theory is proven'. A better statement is: 1

(a) the data are wrong
(b) the theory is proven forever
(c) the theory is false
(d) the theory is strongly supported but remains open to tests

28. A mixture can be separated by sublimation if: 1

(a) only one component sublimes **(b)** both sublime at equal rates
(c) neither component melts **(d)** both melt

29. A student heats 10 g of CaCO3. Only 5.6 g of solid CaO is left. The mass of CO2 evolved is: 1

(a) 4.4 g **(b)** 15.6 g **(c)** 10 g **(d)** 5.6 g

30. An unknown element has isotopes 24 (79 per cent), 25 (10 per cent) and 26 (11 per cent). The average atomic mass is: 1

(a) 25 u **(b)** 24.3 u **(c)** 24 u **(d)** 24.9 u

31. A boat goes 12 km downstream in 1 hour and returns upstream in 2 hours. The average speed for the round trip is: 1

(a) 6 km/h **(b)** 9 km/h **(c)** 8 km/h **(d)** 12 km/h

32. A book is placed on a table. The reaction to the weight of the book is: 1

(a) the pull of the book on the Earth **(b)** friction
(c) the push of the table **(d)** the normal force by the table

33. A block of mass 5 kg is pulled up a smooth incline of length 10 m to a height of 5 m. The work done against gravity is (g = 10): 1

(a) 50 J **(b)** 250 J **(c)** 25 J **(d)** 500 J

34. If the temperature rises from 0 degrees Celsius to 40 degrees Celsius, the speed of sound in air increases from 331 m/s by about (0.6 m/s per degree Celsius): 1

(a) 40 m/s **(b)** 240 m/s **(c)** 6 m/s **(d)** 24 m/s

35. Which of the following could not be a function of the Golgi apparatus? 1

(a) synthesis of ATP **(b)** packaging proteins
(c) formation of lysosomes **(d)** secretion

36. The tissue that produces the enzymes and hormones in glands is: 1

(a) bone **(b)** glandular epithelium
(c) muscle **(d)** squamous epithelium

37. Two animals A and B are in the same order. Which statement is certainly true? 1

(a) They are in the same genus **(b)** They are in the same class and phylum
(c) They are the same species **(d)** They are in the same family

38. A student places a potato piece with an eye in moist soil and another without an eye. The result will be: 1

(a) the one with an eye grows **(b)** the one without an eye grows
(c) both grow **(d)** neither grows

39. A farmer applies too much nitrogen fertiliser. The most likely consequence downstream is: 1

(a) more oxygen **(b)** clear water
(c) algal bloom and fish kill **(d)** more fish

40. Assertion (A): A scientific theory can never be changed. Reason (R): A theory is supported by a large amount of evidence. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

41. **Assertion (A):** Repeating a measurement several times and taking the mean gives a more reliable value than a single reading. 1
Reason (R): Random errors in individual readings tend to cancel out when several readings are averaged.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** A mixture of salt and sand can be separated by dissolving it in water, filtering and evaporating the filtrate. 1
Reason (R): Sand is insoluble in water, whereas salt is soluble in water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** The total mass of the products equals the total mass of the reactants in a chemical reaction carried out in a closed vessel. 1
Reason (R): Atoms are neither created nor destroyed in a chemical reaction; they are only rearranged.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** In Rutherford's experiment most alpha particles passed straight through the gold foil. 1
Reason (R): Most of the space inside an atom is empty.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** The displacement of a body can be zero even though the distance travelled is not zero. 1
Reason (R): Displacement is the shortest straight-line distance from the initial position to the final position, in a given direction.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** Passengers in a bus jerk forward when the bus stops suddenly. 1
Reason (R): The lower part of the body stops with the bus, while the upper part tends to remain in motion because of inertia.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** No work is done by a person who holds a heavy suitcase while standing still. 1
Reason (R): Work is done only when a force produces a displacement of the body in the direction of the force.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Sound cannot travel through a vacuum. 1
Reason (R): Sound is a mechanical wave and needs a material medium for its propagation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** Mitochondria are called the powerhouses of the cell. 1
Reason (R): They release energy from food through cellular respiration and store it in the form of ATP.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** Meristematic tissue is responsible for the growth of plants. 1
Reason (R): Its cells are actively dividing.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**

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